Sahil Singh

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Skills

Languages: Python, TypeScript, JavaScript, Java, SQL

AI & LLM Engineering: Multi-Agent Systems, Deep Agents, LangChain, LangGraph, RAG, GraphRAG, Hybrid Retrieval, Embeddings, MCP, LLM Observability & Evals (LangSmith, Langfuse), Local LLMs

Full-Stack & Distributed Systems: React, Next.js, Node.js, REST APIs, Microservices, API Gateway, Apache Kafka, Redis, WebSockets, Socket.IO, WebRTC, Event-Driven Architecture, System Design

Databases: PostgreSQL, PostGIS, MongoDB, Prisma, Qdrant, Neo4j, Vector Databases

Infrastructure & DevOps: Docker, Kubernetes, Helm, GitHub Actions, CI/CD, AWS (EC2, S3, RDS), Git, BullMQ, Prometheus, Grafana, Loki, Postman

Experience

Codiste PVT LTD Ahmedabad, India
AI Engineer Intern Jul 2026 – Present

- Built and deployed **multi-agent systems** and **Deep Agents** with LangChain and LangGraph across networked machines and local LLM deployments; prototyped **MCP** tool PoCs to evaluate emerging AI frameworks.
- Built a 44-case **LangSmith eval suite** with full tracing and used eval-driven prompt versioning to raise **pass rate from 14% to 95%** and cut hallucinations; reduced **model costs** by migrating agents to a cheaper model after cost and quality benchmarking.
- Cut **agent latency by ∼90%** and **final-output streaming time by ∼70%** by removing redundant LLM/tool calls, parallelizing independent steps, and caching repeated results, enabling faster, production-ready agent responses.
- Resolved **50+ critical bugs** across agent and full-stack code, including state management, and designed a secure **account-sharing system** that grants access without exposing credentials to users, with revoke and reuse policies.

Projects

ChannelIQ - Real-Time Communication Platform Oct 2025 - Jan 2026
GitHub

- Built a Slack-like platform of **5 microservices** (auth, user, chat, media, meeting) behind an **API Gateway** that proxies REST and WebSocket traffic, on React and PostgreSQL.
- Delivered messages in **under 100 ms** using Socket.IO with a **Redis adapter** for horizontal scaling, Redis Pub/Sub fan-out, and **Kafka** write-behind persistence (bulk async PostgreSQL inserts) that keeps the DB off the hot path.
- Secured with JWT cookie auth, Redis sessions/blacklists, and rate limiting; cached profiles and connections for fast reads.
- Integrated LiveKit WebRTC meetings and a media workflow; shipped with Docker, Kubernetes (Helm), GitHub Actions CI/CD, and Prometheus, Grafana, and Loki monitoring.

VendorClause AI - Legal Document Intelligence Platform Jan 2026 - Apr 2026
GitHub

- Built a Next.js legal-document intelligence platform with async ingestion, hybrid vector + graph indexing (Qdrant, Neo4j), and streaming, citation-backed answers.
- Improved retrieval relevance by **∼28%** and cut p95 query latency by **∼40%** through **hybrid retrieval**, graph augmentation, reranking, batching, and caching.
- Engineered **BullMQ ingestion** with deduplication and retry/backoff, Redis Pub/Sub streaming, and **MongoDB checkpoints** for durable LangGraph workflows.
- Orchestrated multi-step agent workflows via MCP with RBAC and audit logs; **profiled tokens, cost, and per-span latency** with Langfuse to tune agent steps.

Civic-Issue - Geo-Spatial Governance Platform Dec 2025 - Jan 2026
GitHub | Live

- Built an offline-first Next.js governance platform with role-based dashboards for workers, engineers, officers, and admins.
- Raised ward-assignment accuracy from **46% to 100%** using **PostGIS point-in-polygon** with spatial indexes, auto-assigning owners with SLA targets.
- Enforced a deterministic workflow (OPEN → VERIFIED) with validation, idempotent retries, and full audit trails.
- Added AI-assisted category suggestions and SLA tracking, cutting manual triage by **60%** in pilot testing.

Education

Parul University Aug 2023 - 2027 (Expected)
B.Tech in Computer Science and Engineering **CGPA: 7.73 / 10**

Relevant Coursework: Data Structures & Algorithms; Operating Systems; DBMS; Computer Networks; OOP.